

ou 2m



Chapter 9 Quiz V2

Name: _____

For full marks, clearly show ALL of your work for every question. diagrams where appropriate. Round your final answer to 3 sig digs. Take a deep breath. Have fun!

1. Monochromatic light produces a third order bright fringe at an angle of 2.8° with a slit separation of 0.25 cm. Include a diagram.

- a. Calculate the wavelength. (3 marks)

$$n=3$$

$$\theta = 2.8$$

$$d = 2.5 \times 10^{-3} \text{ m}$$

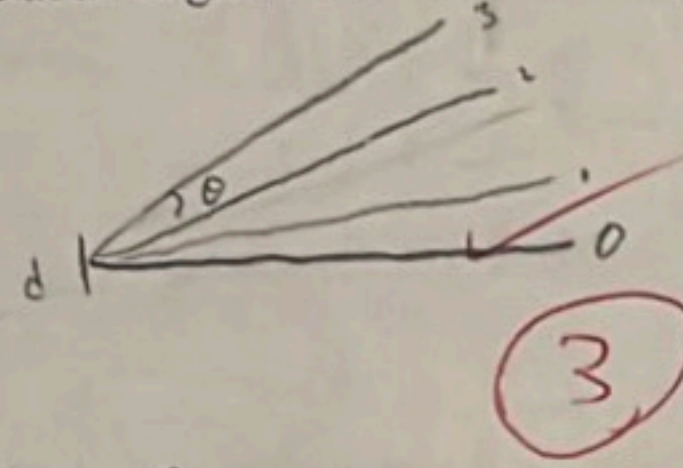
$$\lambda = ?$$

$$d \sin \theta = m \lambda$$

$$\lambda = \frac{d \sin \theta}{m}$$

$$= \frac{(2.5 \times 10^{-3} \text{ m}) \sin 2.8}{3}$$

$$\lambda = 4.07 \times 10^{-5} \text{ m}$$



- b. What angle would the 4th order dark fringe be from the centre point of the sources? (3 marks)

$$n=4$$

$$\lambda = 4.07 \times 10^{-5} \text{ m}$$

$$d = 2.5 \times 10^{-3} \text{ m}$$

$$d \sin \theta = (n - 0.5) \lambda$$

$$\theta = \sin^{-1} \left(\frac{(n - 0.5) \lambda}{d} \right)$$

$$\theta = \sin^{-1} \left(\frac{(3.5) (4.07 \times 10^{-5} \text{ m})}{2.5 \times 10^{-3} \text{ m}} \right)$$

$$\theta = 3.27^\circ$$

2. Light rays pass from air ($n=1.00029$) to sapphire ($n=1.771$) at an angle of 61 degrees with the vertical. a. What is the angle of refraction? Include a diagram in your answer. (2 marks)

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\sin^{-1} \left(\frac{n_1 \sin \theta_1}{n_2} \right) = \theta_2$$

$$\sin^{-1} \left(\frac{(1.00029) \sin 61}{1.771} \right) = \theta_2$$

$$\theta_2 = 29.6^\circ$$

$$\theta_2 \div 30^\circ \times$$



- b. What is the speed of light in a sapphire? (2 marks)

$$n = \frac{c}{v}$$

$$v_1 = \frac{3.00 \times 10^8 \text{ m/s}}{1.00029}$$

$$v_1 = 2.99 \times 10^8 \text{ m/s}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$$

$$v_2 = \frac{v_1 \sin \theta_1}{\sin \theta_2}$$

$$v_2 = 1.69 \times 10^8 \text{ m/s}$$

$$n = \frac{c}{v_2}$$

$$c = (1.771) (1.69 \times 10^8 \text{ m/s})$$

$$c = 2.99 \times 10^8 \text{ m/s}$$

251 cm	246 cm	240 cm	247
250 cm	197 cm	217 cm	253
230 cm	194 cm	230 cm	403
230 cm	207 cm		715
230 cm			



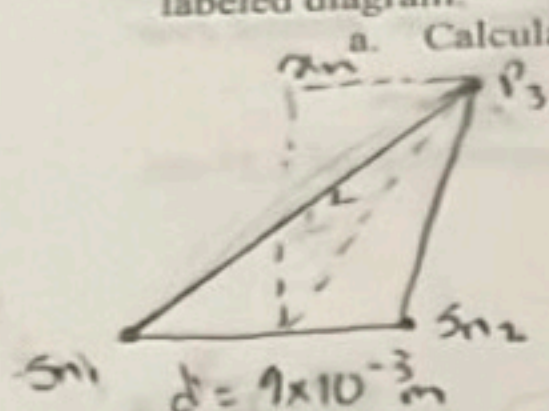
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3. The distance from the bisector to a point P on the third nodal line in a two-point interference pattern is 7.0 cm. The distance from the midpoint between the two sources, which are 0.9 cm apart, to point P is 21 cm. Include a labeled diagram.

a. Calculate the angle for the third nodal line. (3 marks... including 1 for diagram)



$$\begin{aligned} n &= 3 \\ x_n &= 0.07 \text{ m} \\ d &= 9 \times 10^{-3} \text{ m} \\ L &= 0.21 \text{ m} \\ \theta_3 &= ? \end{aligned}$$

$$\sin \theta_3 = (n - \frac{1}{2}) \left(\frac{\lambda}{L} \right)$$

$$\theta_3 = \sin^{-1} \left((2.5) \left(\frac{1.2 \times 10^{-3} \text{ m}}{0.21 \times 10^{-2} \text{ m}} \right) \right)$$

$$\theta_3 = 19.47^\circ$$

$$\theta_3 = 19.5^\circ$$

b. Calculate the wavelength in meters. (2 marks)

$$\lambda = \frac{x}{L} \left(\frac{d}{n - \frac{1}{2}} \right)$$

$$= \left(\frac{0.07 \text{ m}}{0.21 \text{ m}} \right) \left(\frac{9 \times 10^{-3} \text{ m}}{2.5} \right)$$

$$\lambda = 1.2 \times 10^{-3} \text{ m}$$

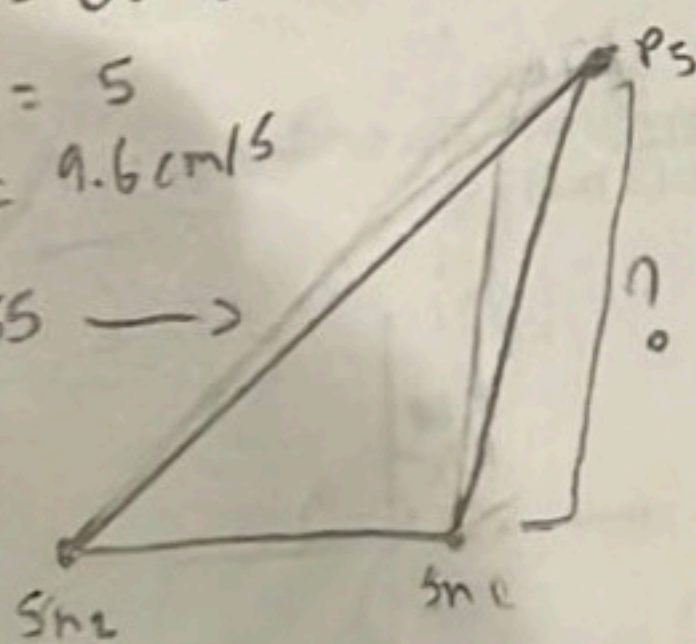
4. Two sources vibrate in phase with a period of 0.154 s. They set up an interference pattern at a point on the 5th nodal line 45.5 cm from one source. Calculate the distance to the second source knowing that the velocity of the wave is 9.6 cm/s. Include a diagram in your answer. (4 marks)

$$T = 0.154 \text{ s}$$

$$n = 5$$

$$v = 9.6 \text{ cm/s}$$

$$S_{n1} = 0.455 \text{ m}$$



$$S_{n2} = ?$$

$$\frac{1}{T} = f$$

$$\frac{1}{0.154 \text{ s}} = f$$

$$6.49 \text{ Hz} = f$$

$$v = \frac{\lambda}{f}$$

$$\lambda = v f$$

$$= (0.096 \text{ m})(6.49 \text{ Hz})$$

$$\lambda = 0.62304 \text{ m}$$

$$9.6$$

$$P_{S1} - P_{S2} = (n - \frac{1}{2}) \lambda$$

$$P_{S1} = (4.5)(0.62304 \text{ m}) = 0.455 \text{ m}$$

$$P_{S1} = 3.26 \text{ m}$$